

EXERCISE 14

Imagine you are an engineer working for a smart building technology company. Your task is to design and simulate a sensor network that monitors and estimates the humidity level in different rooms of a building. Humidity control is crucial for maintaining a comfortable and healthy indoor environment. The sensors you are working with provide humidity measurements, but each sensor's readings are corrupted by a normally distributed noise with a variance of 1.

Some details:

- Each sensor is modeled as a discrete time single-integrator, reflecting the nature of the sensor's measurements over time.
- The sensor readings are corrupted by normally distributed noise with a variance of 1, representing the inherent uncertainty in the measurement process.
- The sensor network needs to be organized in a communication topology that ensures an average consensus. This means that over time, the sensors should collectively converge to a common estimate of the humidity level in the room.
- Choose a suitable communication topology, considering factors such as network robustness, scalability, and efficiency.
- Develop a simulation environment where you can observe the behavior of the sensor network over time.
- Implement algorithms for communication and estimation, ensuring that the sensors' readings are aggregated to achieve an average consensus.
- The final estimated humidity value for the room should be considered as the average of the current sensor estimations. Develop a mechanism to calculate and update this average over time.
- Investigate ways to increase the accuracy of the humidity estimation.
- Define metrics to evaluate the performance of the sensor network. This could include convergence time, stability, and the accuracy of the estimated humidity values.
- Provide clear documentation of your simulation setup, algorithms, and any assumptions made.
- Analyze the simulation results, discussing the impact of the chosen communication topology and any implemented techniques for accuracy improvement.